🟦 UNIT–III

Q1. Explain Windows Operating System Architecture with a neat diagram

How to write (8 marks):

- Definition of Windows OS architecture
- Layered and modular design
- Two execution modes:
  - **User Mode**
  - **Kernel Mode**

User Mode Components

- Applications (legitimate & malware)
- Environment subsystems:
  - Win32 (most important)
  - POSIX
  - .NET CLR
- System processes:
  - csrss.exe
  - winlogon.exe
  - services.exe
- DLLs:
  - kernel32.dll
  - user32.dll
  - advapi32.dll

Kernel Mode Components

- Windows Executive:
  - Process Manager

    - Memory Manager
    - I/O Manager
    - Security Reference Monitor
    - Object Manager
    - Configuration Manager
- Kernel (scheduling, interrupts)
- HAL (hardware abstraction)
- Device drivers

Forensic relevance

- Kernel rootkits
- Memory dump analysis
- Process & registry artifacts

Neat diagram + conclusion

Q2. Explain FAT File System Architecture and its forensic significance

- Definition of FAT file system
- Usage in USBs, SD cards, removable media

FAT Structure

1. Boot Sector (VBR)
    - BPB
    - FAT type
    - Cluster size
2. FAT Tables (FAT1 & FAT2)
    - Cluster mapping
    - Redundancy
3. Root Directory

    - Fixed (FAT12/16)
    - Data area (FAT32)
4. Data Area
    - Actual file content

Cluster chains

- Linked list storage
- Orphan clusters

Forensic significance

- Deleted file recovery (0xE5)
- Timestamp analysis
- Slack space analysis
- Simplicity aids carving
- Common in cybercrime devices

Conclusion

Q3. Explain NTFS File System Architecture and the role of MFT

- Definition of NTFS
- Default Windows file system

NTFS Layout

1. Boot sector
2. Master File Table (MFT)
3. NTFS system files:
    - $MFT
    - $MFTMirr
    - $Bitmap
    - $LogFile

    - $Secure

4. Data area

MFT Details

- Every file = one MFT entry
- Entry size ≈ 1024 bytes

Important MFT Attributes

- $STANDARD_INFORMATION
- $FILE_NAME
- $DATA
- $INDEX_ROOT / $INDEX_ALLOCATION

Forensic significance

- Metadata survives deletion
- Multiple timestamps
- Journaling evidence
- Deleted file recovery

Conclusion

Q4. Explain how to recreate FAT and NTFS partitions and their forensic importance

Recreating FAT Partitions

- Analyze VBR parameters
- Examine FAT tables
- Recover root directory entries
- Manual carving
- Tool-based recovery (TestDisk, Autopsy)

Forensic importance (FAT)

- Recover deleted USB partitions

- Restore cluster chains
- Timeline reconstruction

Recreating NTFS Partitions

- Analyze NTFS boot sector
- Locate MFT using signature "FILE"
- Use $MFTMirr
- Analyze $Bitmap & $LogFile
- Rebuild partition table using tools

Forensic importance (NTFS)

- Restore formatted partitions
- Recover MFT metadata
- Detect anti-forensics

Conclusion

Q5. Explain Windows Registry, its structure, and forensic importance

- Definition of Windows Registry
- Central configuration database

Registry Structure

- Hives, keys, subkeys, values

Root Hives

- HKLM
- HKCU
- HKCR
- HKU
- HKCC

Physical Registry Files

- SYSTEM
- SOFTWARE
- SAM
- NTUSER.DAT
- USRCLASS.DAT

Registry Value Types

- REG_SZ
- REG_DWORD
- REG_BINARY
- REG_MULTI_SZ

Forensic importance

- User activity traces
- USB history
- Application usage
- Malware persistence
- Timeline analysis

Conclusion

Q6. Describe important Windows Registry artifacts used in forensic investigations

Key Artifacts

- RecentDocs
- RunMRU
- OpenSaveMRU
- ShellBags
- TypedURLs / TypedPaths
- UserAssist (ROT13)

- USBSTOR
- Installed applications
- Network profiles
- Prefetch correlation

Forensic use

- Proves file access
- Program execution
- Device usage
- User behavior reconstruction

Conclusion

Q7. Explain Windows Event Logs, their structure, and forensic importance

Types of Logs

- System
- Security
- Application
- Setup
- Forwarded events

Event Log Structure

- Header (record ID, timestamp)
- Event data (SID, PID, action)
- XML metadata (EVTX)

Forensic importance

- Authentication tracking
- Malware detection
- Timeline reconstruction

- Boot/shutdown events
- Evidence correlation

Conclusion

Q8. Explain EVT and EVTX log formats and forensic analysis

EVT (Legacy)

- Binary
- Limited size
- No XML
- Prone to corruption

EVTX (Modern)

- XML-based
- Chunk structure
- Checksums
- Better recovery

Forensic Analysis

- Extraction paths
- Hash validation
- Viewing tools
- Key event IDs
- Tampering detection
- Artifact correlation

Conclusion

Q9. Explain analysis of third-party Windows application logs with examples

Types of Logs

- Browser logs (Chrome, Firefox)
- Messaging apps (Skype, WhatsApp)
- Antivirus logs
- Cloud application logs
- Office logs

Analysis Techniques

- Locate log paths
- SQLite database analysis
- Timestamp analysis
- Keyword searching
- Deleted log recovery

Examples

- Browser history
- Cloud upload traces
- Chat evidence

Conclusion

Q10. Explain Prefetch files, their structure, and forensic significance

- Definition of Prefetch files
- Location: C:\Windows\Prefetch

Purpose

- Improve application load time

Structure

- Header
- Execution info
- File metrics

- Trace chains
- Volume information

Forensic significance

- Program execution proof
- Run count
- Timestamp correlation
- Deleted prefetch recovery
- Anti-forensic detection

Conclusion

Q11. Explain analysis of Unallocated Space in Windows file systems

- Definition of unallocated space

Sources of data

- Deleted files
- Fragmented files
- Old partitions
- Partial overwrites

Analysis Techniques

- File carving
- Keyword search
- Metadata recovery
- Entropy analysis

Forensic importance

- Evidence recovery
- Hidden data detection
- Timeline reconstruction

- Anti-forensics detection

Challenges

- SSD TRIM
- Fragmentation

Conclusion

Q12. Explain analysis of Program Execution Artifacts in Windows

Key Artifacts

- Prefetch files
- Event logs (4688, 7045)
- ShimCache
- AmCache
- UserAssist
- MUICache
- Jump Lists
- Registry Run keys

Correlation

- Prefetch + Event Logs
- Registry + MFT timestamps

Forensic importance

- Confirms execution
- Malware detection
- User behavior analysis

Conclusion
